

Introduction

This model is a specific 22.1M parameters Multimodal Neural Network design to predict the future 3s trajectory of a vehicle. It processes visual data from two images with a temporal gap of 0.1s, along with vehicle dynamics (speed, acceleration, and turn rate) derived from GPS data. The model outputs a sequence of 12 displacement vectors representing the future 3s vehicle's path.

1 Model Architecture

1. Image Processing Branch (Convolutional Backbone with Cross-Processing for Two Images):

- **Layer 1:** Convolutional layer with 64 filters, kernel size 3×3 , ReLU activation, input shape $(480, 270, 3)$, applied to the first image.
- **Layer 2:** Batch normalization.
- **Layer 3:** Max-pooling layer with 2×2 pool size.
- **Layer 4:** Convolutional layer with 128 filters, kernel size 3×3 , ReLU activation.
- **Layer 5:** Batch normalization.
- **Layer 6:** Convolutional layer with 64 filters, kernel size 3×3 , ReLU activation, input shape $(480, 270, 3)$, applied to the second image (shared weights with Layers 1–5).
- **Layer 7:** Batch normalization.
- **Layer 8:** Max-pooling layer with 2×2 pool size.
- **Layer 9:** Convolutional layer with 128 filters, kernel size 3×3 , ReLU activation.
- **Layer 10:** Batch normalization.
- **Layer 11:** Cross-attention layer, where feature maps from the first image (from Layer 5, $240 \times 135 \times 128$) attend to feature maps from the second image (from Layer 10, $240 \times 135 \times 128$), and vice versa, producing enhanced feature maps for each image ($240 \times 135 \times 128$).
- **Layer 12:** Max-pooling layer with 2×2 pool size, applied to the first image's enhanced feature maps.
- **Layer 13:** Convolutional layer with 256 filters, kernel size 3×3 , ReLU activation.
- **Layer 14:** Batch normalization.
- **Layer 15:** Convolutional layer with 256 filters, kernel size 3×3 , ReLU activation.
- **Layer 16:** Batch normalization.
- **Layer 17:** Max-pooling layer with 2×2 pool size, applied to the second image's enhanced feature maps.
- **Layer 18:** Convolutional layer with 256 filters, kernel size 3×3 , ReLU activation.
- **Layer 19:** Batch normalization.
- **Layer 20:** Convolutional layer with 256 filters, kernel size 3×3 , ReLU activation.
- **Layer 21:** Batch normalization.
- **Layer 22:** Cross-attention layer, where feature maps from the first image (from Layer 16, $120 \times 67 \times 256$) attend to feature maps from the second image (from Layer 21, $120 \times 67 \times 256$), and vice versa, producing enhanced feature maps ($120 \times 67 \times 256$).
- **Layer 23:** Max-pooling layer with 2×2 pool size, applied to the first image's enhanced feature maps.
- **Layer 24:** Convolutional layer with 512 filters, kernel size 3×3 , ReLU activation.
- **Layer 25:** Batch normalization.
- **Layer 26:** Convolutional layer with 512 filters, kernel size 3×3 , ReLU activation.
- **Layer 27:** Batch normalization.
- **Layer 28:** Max-pooling layer with 2×2 pool size, applied to the second image's enhanced feature maps.

- **Layer 29:** Convolutional layer with 512 filters, kernel size 3×3 , ReLU activation.
 - **Layer 30:** Batch normalization.
 - **Layer 31:** Convolutional layer with 512 filters, kernel size 3×3 , ReLU activation.
 - **Layer 32:** Batch normalization.
 - **Layer 33:** Concatenation layer to merge the feature maps from both images (e.g., combining two $60 \times 33 \times 512$ feature maps into $60 \times 33 \times 1024$).
 - **Layer 34:** Convolutional layer with 1024 filters, kernel size 3×3 , ReLU activation, applied to the concatenated feature maps.
 - **Layer 35:** Global Average Pooling layer to vectorize the feature maps from the previous layer into a 1024-dimensional vector.
2. **Dynamic Input Processing Branch (Speed, Acceleration, Turn Rate):**
- **Layer 36:** Dense layer with 32 units, ReLU activation, input shape (3).
 - **Layer 37:** Batch normalization.
3. **Feature Fusion:**
- **Layer 38:** Concatenation layer to merge the vectorized image features (from Layer 35) and the processed dynamic features (from Layer 37), producing a fused feature vector of size 1056.
4. **Fully Connected Layers:**
- **Layer 39:** Dense layer with 2048 units, ReLU activation.
 - **Layer 40:** Dropout layer with 0.5 dropout rate.
 - **Layer 41:** Dense layer with 512 units, ReLU activation.
 - **Layer 42:** Dropout layer with 0.5 dropout rate.
 - **Layer 43:** Dense layer with 128 units, ReLU activation.
5. **Output Layer:**
- **Layer 44:** Dense layer with 24 units, linear activation.
 - **Layer 45:** Reshape layer to format the output as a tensor of shape (12, 2).